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Long-term periodontal maintenance in a patient with Papillon–Lefèvre syndrome: An 8-year case report

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【Abstract】 【Background】 Papillon–Lefèvre syndrome (PLS) is an autosomal recessive disorder with a prevalence of approximately 1–4 cases per one million individuals. This condition frequently leads to premature loss of both the primary and the permanent teeth. PLS is primarily caused by mutations in the cathepsin C (CTSC) gene, resulting in the loss of CTSC function, which disrupts immune regulation and triggers severe inflammatory responses. Maintaining and preserving teeth in these patients therefore presents a significant challenge for periodontists. **【Case Presentation】** The patient received multidisciplinary treatment involving periodontics, oral and maxillofacial surgery, and prosthodontics. Hopeless teeth were extracted, removable dentures were provided, and supportive periodontal maintenance was performed over an 8-year period. Adjunctive systemic antibiotics were used during active treatment and selected maintenance visits because of severe inflammation and multiple sites with suppuration, but this approach was not designed as a standardized long-term antibiotic protocol.

【Results】 At the latest follow-up, 22 natural teeth were retained. The bleeding on probing rate decreased from 71% at baseline to 18% in March 2025, the

proportion of sites with probing depth ≥ 4 mm decreased from 54% to 27%, and the Plaque Index improved from 2–3 to 1–2. Clinical attachment level was not consistently recorded and therefore could not be analyzed longitudinally.

【Conclusion】 This single case suggests that early diagnosis, extraction of hopeless teeth, individualized multidisciplinary periodontal–prosthodontic management, regular supportive periodontal care, sustained plaque control, and strong family cooperation may help prolong the retention of functional natural teeth in selected patients with Papillon–Lefèvre syndrome. Further longitudinal studies with standardized periodontal, microbiological, and antimicrobial-resistance monitoring are needed.

【Key words】 Papillon-Lefèvre syndrome; cathepsin C; gene mutation; periodontal treatment; case report

Background

Papillon–Lefèvre syndrome (PLS) was first reported by Papillon and Lefèvre in 1924^[1]. This condition is an autosomal recessive disorder with a prevalence of approximately 1–4 cases per million individuals, and approximately one-third of the patients have a family history of consanguineous marriage^[2]. Its clinical features include rapid early destruction of periodontal tissues and hyperkeratosis of the skin on the elbows, knees, palms, and soles. The incidence is equal in males and females, with no significant differences across ethnicities^[3]. This condition often leads to the premature loss of both primary and permanent teeth. PLS is primarily caused by mutations in the *CTSC* gene. These mutations may alter the structure of the peptide chain, affect its folding and assembly, hinder the formation of functional *CTSC* protein, modify the structure of the *CTSC* protein, reduce or even inhibit its activity, impact immune responses, and result in the loss of *CTSC* function. This leads to a disease phenotype, which causes immune dysregulation and triggers severe inflammatory responses^[4]. *CTSC* is highly expressed in immune cells such as neutrophils, cytotoxic lymphocytes, natural killer cells, macrophages within the alveolar bone, and mast cells. *CTSC* can activate neutrophil-derived serine proteases and granzymes, thereby influencing immune responses^[5]. Immune dysregulation caused by

CTSC gene mutations increases inflammatory susceptibility in patients with PLS and increases their susceptibility to severe inflammatory diseases^[6]. In 2017, the Department of Periodontics at the Affiliated Stomatological Hospital of Nanchang University admitted a patient with this syndrome. Through a multidisciplinary approach that combined periodontal, surgical, and prosthodontic treatments, most of the patient's permanent teeth were successfully preserved. We report the case in this paper.

Case presentation

Initial Presentation

A 10-year-old male patient presented to the Department of Periodontics at the Affiliated Stomatological Hospital of Nanchang University on June 28, 2017. The chief complaint was looseness of tooth starting at age 2, accompanied by the loss of some teeth, difficulty with normal chewing, and recurrent gum swelling and pain. Most of the teeth gradually loosened and were subsequently lost. The patient's parents reported that after 8 months of age, hyperkeratosis, fissuring, and scaling appeared on the palms of both hands, the soles of both feet, elbows, and knees, which worsened during the winter. The patient showed no obvious abnormalities in growth or development and was of normal intelligence. After the age of 6, he was able to brush his teeth twice daily, but his oral hygiene remained poor.

Physical Examination

Growth, development and intelligence were all normal. Hyperkeratosis, fissuring, and scaling were observed on the palms of both hands, soles of both feet, elbows, and knees, with variable thickness of the keratin layer, which presented an "ichthyosiform" appearance (Figure 3). The skin lesions on both knees extended to the groin area, while mild scaling was noted in the preauricular area. No significant abnormalities were observed in the fingernails, toenails, or hair, and no significant hyperhidrosis or fetid sweat was noted.

Family History

No history of consanguineous marriage among the parents was noted. The patient had three immediate relatives (father, mother, and a younger brother). Periodontal and systemic examination revealed that neither his parents nor his brother exhibited periodontitis or palmoplantar keratoderma.

Oral Examination

The facial features of the patient were symmetrical, with a mouth opening of approximately three fingerbreadths, and the patient was in the permanent dentition stage. Oral hygiene was poor, and dental calculus was visible (++). The gingiva in the anterior region was red, swollen and exhibited a soft consistency. The bleeding on probing (BOP) rate was 71%, and the probing depth (PD) ranged from approximately 3 to 10 mm. Teeth 16, 12, 11, 21, 22, 26, 32, 42, and 46 showed grade III mobility, whereas tooth 36 showed grade II mobility. Teeth 31 and 41 were missing, and the edentulous alveolar ridge in the mandibular anterior region was relatively flat. No significant abnormalities were detected in tooth development or caries (Figure 1). Radiographic examination revealed extensive generalized alveolar bone loss, as bone resorption reached the periapical region in the anterior teeth; however, no significant bone loss was observed around teeth with open apices or in unerupted teeth (Figure 2).

Other Tests

Genetic testing of the patient revealed compound heterozygous mutations in the CTSC gene, specifically c.800T>C in exon 6 and c.1015C>T in exon 7. Serological examination at the initial visit revealed no Aa infection..Aa was not re-tested during subsequent follow-up.



Figure 1: Intraoral image of the patient at the initial visit in June 2017. Poor oral hygiene with gingival redness and swelling.

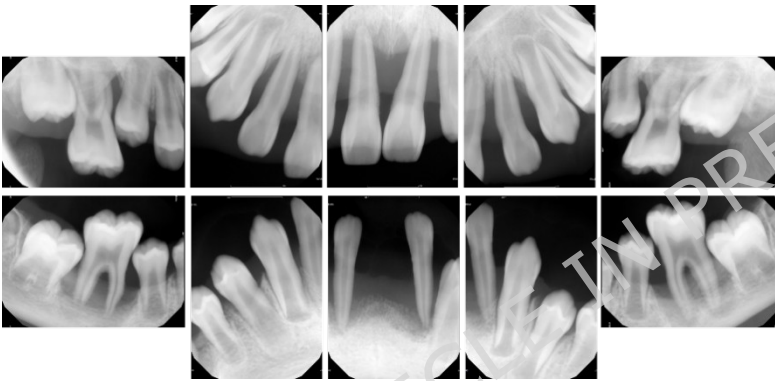


Figure 2: Full-mouth periapical radiograph obtained at the initial visit. Significant alveolar bone resorption is observed in the anterior tooth region and the second molars.



Figure 3: Skin of the palms, soles, elbows, and knees at the initial visit. All exhibit hyperkeratosis.

Diagnosis

On the basis of early-onset palmoplantar hyperkeratosis and severe periodontal tissue destruction, the patient received a clinical diagnosis of PLS. The diagnosis was confirmed by the presence of compound heterozygous mutations in the CTSC gene, which were found in both parents who were asymptomatic carriers.

Treatment

Oral hygiene instructions were provided to the patient. Supragingival scaling was performed, followed by subgingival scaling and root planing at sites with probing depth ≥ 4 mm. Teeth 11, 21, 32, 42, 16, 26, 36, and 46 were extracted because of severe periodontal destruction and poor prognosis. Removable dentures were fitted three months after extractions to restore mastication, speech, and facial appearance.

From June 2017 to July 2025, supportive periodontal maintenance therapy was performed in the Department of Periodontics at intervals of approximately 3 to 6 months. During the later follow-up period, maintenance intervals were shortened to approximately 3 to 4 months when feasible. The patient was instructed to brush regularly, use dental floss as an aid, and rinse with 0.12% chlorhexidine solution after meals. Follow-up communication was maintained through telephone and WeChat, and immediate consultations were arranged when discomfort occurred. Adjunctive systemic antibiotics were prescribed in combination with mechanical periodontal therapy because, at the initial presentation, subgingival instrumentation alone was considered insufficient to control the severe inflammatory condition, and suppuration was present at multiple periodontal sites. Amoxicillin combined with metronidazole was selected based on previous reports of systemic adjunctive therapy in patients with Papillon–Lefèvre syndrome. Amoxicillin is administered every 8 hours at a dose of 500 mg each time, and metronidazole is administered every 8 hours at a dose of 250 mg each time, for a total duration of 7 days^[7]. The patient and family did not report gastrointestinal discomfort, allergy, or other obvious antibiotic-related adverse reactions during follow-up. Nevertheless, adverse events were not recorded using a standardized surveillance form, and antimicrobial-resistance testing was not performed. The use of systemic antibiotics in this case should be interpreted with

caution and should not be regarded as a recommended standard treatment regimen for patients with Papillon-Lefevre syndrome.

Timeline and Follow-up Outcomes

The longitudinal clinical findings are summarized in Table 1.

Table 1, CARE timeline

Time point/period	Dentition status	Periodontal findings	Maintenance/prosthetic/radiographic findings	Microbiology/CAL	Key clinical decision / rationale
June 2017 (initial visit)	17 natural teeth; 2 missing teeth (31, 41). Teeth 16, 12, 11, 21, 32, and 42 showed grade III mobility; teeth 22, 36, and 46 showed grade II mobility.	BOP(+): 71%; PD $\geq$ 4 mm: 54%; PLI: 2–3.	Extensive generalized alveolar bone resorption; no significant bone loss around teeth with unformed apices or tooth germs.	Aa: not detected at baseline. CAL: not consistently recorded.	Extract hopeless teeth to reduce inflammatory burden and improve hygiene around remaining teeth
November 2017	12 natural teeth; 9 missing teeth (16, 11, 21, 26, 32, 31, 41, 42, and 46). Teeth 12, 22, and 36 showed grade II mobility.	BOP(+): 17%; PD $\geq$ 4 mm: 25%; PLI: not recorded.	Initial periodontal therapy completed; missing teeth restored with a clasp-retained removable denture.	Aa: not re-tested. CAL: not consistently recorded.	Provide removable denture to restore mastication, speech, and appearance
February 2018–August 2022	18 natural teeth; 10 missing teeth (16, 11, 21, 26, 36, 32, 31, 41, 42, and 46). Teeth 12 and 22 showed grade II mobility.	July 2020: BOP(+): 39%; PD $\geq$ 4 mm: 34%; PLI: 1–2.	Supportive periodontal maintenance every 3–6 months over 18 visits. Denture relining in January 2019 and new mandibular removable denture in July 2020. January 2022 radiograph showed stable mild bone loss in most remaining teeth and complete apical closure.	Aa: not re-tested. CAL: not consistently recorded.	Continue supportive periodontal care during mixed-to-permanent dentition transition
October 2022–July 2025	22 natural teeth; 10 missing teeth. Teeth 12 and 22 showed grade II mobility.	March 2025: BOP(+): 18%; PD $\geq$ 4 mm: 27%; PLI: 1–2.	Supportive periodontal maintenance every 3–4 months over 10 visits. Restorative treatment was provided for teeth 12, 22, 24, 37, and 47; pericoronal irrigation was performed for tooth 48. March 2025 radiograph showed mild bone loss in most remaining teeth	Aa: not re-tested. CAL: not consistently recorded.	Preserve teeth 12 and 22 under close maintenance because of patient's aesthetic preference

			and vertically erupting third molars without significant alveolar bone loss.		
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Note*:In clinical examinations, a visual inspection combined with probing method was employed ^[8]

Clinical Outcomes

After initial therapy and long-term maintenance, periodontal inflammation decreased. The BOP rate decreased from 71% at baseline to 18% in March 2025. The proportion of sites with PD ≥ 4 mm decreased from 54% to 27%. The Plaque Index improved from 2–3 to 1–2. No spontaneous loss of natural teeth was recorded after the initial treatment phase. The number of retained natural teeth increased during follow-up because additional permanent teeth erupted during adolescence. At the latest follow-up, the patient retained 22 natural teeth. The removable dentures were comfortable to wear, and the patient reported that his natural teeth were stable. The patient's parents accompanied him for a total of 47 visits over the 8-year follow-up period, reflecting strong family support and adherence to maintenance care. After receiving oral hygiene instructions, the patient brushed his teeth three times per day and used dental floss as an aid. The patient and family maintained close contact with the clinicians and attended follow-up visits when possible. Full-mouth periodontal probing was performed in detail on three occasions by the treating clinicians during the 8-year treatment period. However, because the patient was treated from the mixed dentition stage to the permanent dentition stage, and because several teeth erupted, were extracted, or changed in occlusal and prosthodontic status during growth, clinical attachment level measurements were not recorded in a sufficiently standardized and continuous manner for reliable longitudinal analysis. To monitor periodontal destruction more intuitively during craniofacial growth and dentition transition, panoramic radiographs were obtained approximately every six months to evaluate changes in alveolar bone height and progression of bone loss. Microbiological evaluation was also limited. Aa was not detected at the initial visit, but the patient and family declined repeated microbiological testing during subsequent follow-up. Therefore, this case cannot determine whether microbiological stability, changes in bacterial load, or bacterial eradication contributed to disease control. The absence of longitudinal CAL and

microbiological data represents an important limitation of this retrospective case report.



Figure 4: Panoramic radiograph from July 2018 showing horizontal alveolar bone loss that reached the apical third root of teeth 12 and 22, the presence of tooth germs in teeth 18, 28, 38, and 48, mild bone loss in the remaining teeth; the apical foramen was not yet closed.



Figure 5: Panoramic radiograph from January 2022 showing horizontal alveolar bone loss that reached the apical third root of teeth 12 and 22, the presence of

tooth germs in teeth 18, 28, 38, and 48, mild bone loss in the remaining teeth, and complete closure of the apical foramen.



Figure 6: Panoramic radiograph from March 2025. Compared with previous radiographs, no obvious generalized progression of alveolar bone loss was observed in most retained teeth.



Figure 7: The patient's palms, soles, knees, and elbows still show hyperkeratosis and scaling.



Figure 8: Intraoral photograph obtained in March 2025, The patient is satisfied with the effect of removable denture restoration for missing teeth.

Patient Perspective

The patient and his family expressed satisfaction with the long-term treatment outcome. Before referral to our department, the patient had experienced premature loss of primary teeth and progressive impairment of masticatory function. At the age of 10 years, chewing function was already

substantially affected. After multidisciplinary periodontal, surgical, and prosthodontic management, the patient reported improvement in mastication, speech, facial appearance, and comfort while wearing removable dentures.

During treatment, the clinicians also considered the patient's own preferences. For example, although teeth 12 and 22 had limited periodontal prognosis from a strictly clinical perspective, the patient strongly wished to retain them because he considered them important for facial appearance. After discussion, the treatment team respected this preference and attempted to preserve these teeth under close maintenance. This shared decision-making process improved the patient's understanding of the treatment plan and strengthened his long-term compliance. The patient and family believed that preserving functional and aesthetically important natural teeth had a positive impact on his quality of life during adolescence.

Discussion

1. Diagnostic considerations and comparison with previous reports

The pathogenesis of PLS, which is an autosomal recessive disorder, involves mutations in the *CTSC* gene located at 11q14.2. PLS often leads to the premature loss of primary and permanent teeth and may be accompanied by palmoplantar hyperkeratosis, with skin lesions typically worsening during the winter^[9]. However, in some cases, the third molars may be spared from periodontal destruction^[10]. In addition to these two classic manifestations, PLS has been reported to be associated with various systemic diseases, among which recurrent infections are relatively common and may present as pyoderma and liver abscesses, among others. In recent decades, cases of PLS complicated by xanthogranulomatous pyelonephritis and asymptomatic dural ectopic calcification have been reported^[11]. Furthermore, PLS sometimes requires

differentiation from Haim–Munk syndrome (HMS). HMS is also an extremely rare autosomal recessive disorder associated with *CTSC* gene mutations. In addition to palmoplantar hyperkeratosis and periodontal tissue destruction, HMS is characterized by acro-osteolysis of the distal phalanges, arachnodactyly (spider-like fingers), and characteristic curved nails and pes planus ^[12].

In this study, sequencing of exons 1–7 of the *CTSC* gene in the patient's family revealed that the patient's parents each carried heterozygous mutations of c.800T>C and c.1015C>T. Other studies have indicated that periodontal inflammation in patients with PLS may peak during adolescence and subsequently diminish with age ^[13].

Reports on the long-term follow-up of patients with PLS remain scarce. Most patients are diagnosed at preschool or school age (2–11 years) and receive periodontal treatment, systemic antibiotics, and regular maintenance visits. However, approximately 90% of patients ultimately lose all their permanent teeth, and often, only some third molars are preserved ^[14,15]. This finding indicates that, even with professional treatment and follow-up, the long-term prognosis for the dentition of patients with PLS remains generally poor.

Currently, no universally accepted or definitively effective treatment protocol has been established for PLS. Clinical management often involves conventional scaling/supragingival scaling, systemic antibiotics, and the extraction of teeth that cannot be saved, but the overall therapeutic outcomes show considerable individual variation^[16]. Aldevina C. de Freitas et al. ^[17] conducted a 14-year follow-up on a 9-year-old Brazilian girl with PLS. Although she received intensive combined periodontal and antibiotic therapy in the early stages, she lost all her permanent teeth within 4 years and had to be rehabilitated with complete removable dentures. De Vree et al. ^[18] conducted a 15-year follow-up on a pair of siblings with PLS and suggested that the presence of an edentulous period during the transition from primary to permanent dentition may lead to more stable preservation of natural teeth. Notably, the female sibling retained a significant number of permanent teeth in the later stages of follow-up, and Aa was no longer detected, whereas the male sibling became edentulous, and

substantial Aa was still detectable in his serum. This disparity may be related to metronidazole resistance in the male or differences in individual host responses and compliance. In the present case, Serological examination at the initial visit revealed no evidence of *Aggregatibacter actinomycetemcomitans* (Aa) infection. Aa was not re-tested during subsequent follow-up. Aa was tested only at the initial visit and was not monitored longitudinally; therefore, no conclusion can be drawn regarding microbiological eradication or changes in bacterial load.

2. Possible reasons for favourable tooth retention

The relatively favorable tooth retention observed in this patient should be interpreted as the result of multiple interacting factors. First, teeth with severe mobility and hopeless prognosis were extracted during the early treatment phase, which may have reduced the inflammatory burden and facilitated plaque control around the remaining teeth. Second, supportive periodontal care was maintained for 8 years, and the patient's family showed exceptional cooperation, accompanying him to 47 visits and maintaining close communication with the clinicians. Third, the patient gradually developed effective self-care behavior, including regular toothbrushing, flossing, chlorhexidine rinsing, and timely consultation when symptoms occurred. Fourth, eruption of additional permanent teeth during adolescence contributed to the increased number of retained natural teeth during follow-up. Fifth, prosthodontic rehabilitation helped restore mastication, speech, and facial appearance, which may have improved the patient's motivation and adherence.

The shared decision-making process may also have contributed to long-term adherence. In this case, the patient strongly wished to retain teeth 12 and 22 because of their aesthetic importance, although their periodontal prognosis was limited. The treatment team respected this preference and attempted to preserve these teeth under close monitoring. This approach helped the patient feel involved in treatment decisions and may have strengthened his trust in the clinicians. For adolescent patients with rare diseases and long treatment courses, patient-centered communication may be especially important because treatment goals involve not only tooth retention but also appearance, function, psychological development, and quality of life.

The multidisciplinary nature of the treatment was another important factor. Periodontal therapy controlled inflammation, oral and maxillofacial surgery removed teeth with hopeless prognosis, and prosthodontic rehabilitation restored missing teeth and supported oral function during growth. For a child entering adolescence, such combined management is clinically meaningful. The goal should not be limited to extracting teeth and providing prosthetic replacement, but should include preserving strategically and aesthetically important teeth when feasible, maintaining function during growth, and supporting the patient's confidence and daily quality of life.

3.Limitations and clinical implications

This case also has important limitations. First, it is a single retrospective case report and cannot establish causality or define an optimal treatment protocol. Second, full-mouth CAL measurements were not consistently recorded in a standardized manner throughout the 8-year period. Although BOP, PD ≥ 4 mm, PLI, clinical mobility, tooth retention, and radiographic bone height were followed, these indicators cannot fully replace longitudinal CAL data. Third, microbiological testing was performed only at baseline, and Aa was not re-tested because the patient and family declined further testing. Therefore, no conclusion can be drawn regarding changes in microbial load, bacterial eradication, or the microbiological basis of disease control. Fourth, antibiotic-related adverse events and antimicrobial resistance were not systematically monitored. Fifth, because the patient underwent growth and dentition transition from childhood to adolescence, changes in tooth number and periodontal status should be interpreted in the context of eruption, extraction, prosthodontic adaptation, and craniofacial development.

Despite these limitations, this case provides useful clinical insight. It suggests that in selected patients with Papillon–Lefèvre syndrome, the goal of treatment may be more conservative and functional than complete early extraction. Early recognition, interdisciplinary treatment, removal of hopeless teeth, regular supportive periodontal therapy, family cooperation, plaque control, prosthodontic rehabilitation, and respect for patient preferences may help prolong the retention of functional natural teeth. The main value of this report is not to demonstrate a standardized drug regimen, but to document a long-term,

patient-centered management process in a rare and difficult periodontal condition.

Conclusion

This 8-year case report describes a patient with Papillon–Lefèvre syndrome who retained 22 natural teeth after multidisciplinary periodontal, surgical, and prosthodontic management. The favorable outcome was likely related to early diagnosis after referral, extraction of hopeless teeth, regular supportive periodontal therapy, prosthodontic rehabilitation, sustained plaque control, strong family cooperation, and shared decision-making. The repeated use of systemic antibiotics should be interpreted cautiously and should not be generalized as a standard long-term maintenance protocol. Future studies should include standardized CAL recording, longitudinal microbiological monitoring, adverse-event documentation, and antimicrobial-resistance assessment to better define safe and effective long-term management strategies for patients with Papillon–Lefèvre syndrome.

Declarations

Abbreviations

PLS: Papillon-Lefèvre syndrome;

CAL: clinical attachment level;

BOP: bleeding on probing;

PD: probing depth;

CTSC: cathepsin C.

Ethics approval and consent to participate

This retrospective single - case report used routine clinical data and was conducted in accordance with the principles of the Declaration of Helsinki.

The Ethics Committee of the Affiliated Stomatological Hospital of Nanchang University reviewed the study and confirmed that formal ethics approval was waived for this type of retrospective single - case report involving de - identified routine clinical data (waiver number: [2025(093)]). Written informed consent for treatment and for the academic use of de-identified clinical information, photographs, radiographs, and genetic testing results was obtained from the patient' s mother when the patient was under 18 years of age. After the patient reached 18 years of age in 2025, written informed consent was obtained again from the patient himself.

Consent for publication

Written informed consent for publication of anonymized clinical data, images, and radiographs was obtained from the patient himself in 2025 (he was over 18 years of age at that time). The consent form also includes the mother' s signature from the period when the patient was a minor. A copy of the signed consent form has been uploaded as a separate file under “ Related files” . All authors are aware of the submission and agree to publish this article.

Availability of data and materials

All relevant clinical data supporting the findings of this study are included within the manuscript. Raw clinical records are not publicly available due to patient privacy restrictions but can be requested from the corresponding author upon reasonable request.

Competing Interests

The authors declare that they have no competing interests.

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Authors' contributions

Tingting Hu wrote the main manuscript. Fang Ye and Tingting Hu provided comprehensive clinical guidance. Xiaoyan Zou contributed partial diagnostic insights. Yuebo Wu compiled the relevant clinical images. Tingting Hu and Fang Ye edited the final version. All authors reviewed and approved the final manuscript.

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Clinical trial number

not applicable

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